

SHEET 1

Write and balance the following equations:

1. aluminium hydroxide + sulfuric acid $\rightarrow$ aluminium sulfate + water



2. zinc oxide + nitric acid $\rightarrow$ zinc nitrate + water



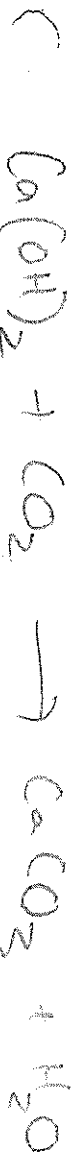
3. iron + oxygen gas $\rightarrow$ iron (III) oxide



4. magnesium + nitric acid $\rightarrow$ magnesium nitrate + hydrogen gas



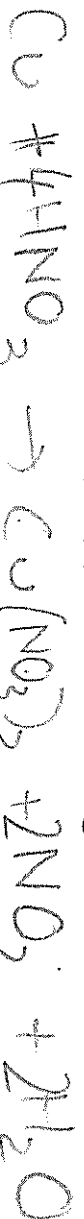
5. calcium hydroxide + carbon dioxide $\rightarrow$ calcium carbonate + water



6. potassium + water $\rightarrow$ potassium hydroxide + hydrogen gas



7. copper + nitric acid $\rightarrow$ copper (II) nitrate + nitrogen dioxide + water



8. magnesium hydroxide + hydrochloric acid $\rightarrow$ magnesium chloride + water



9. zinc hydroxide + hydrochloric acid $\rightarrow$ zinc chloride + water



10. sodium hydrogen carbonate $\xrightarrow{\text{heat}}$ sodium carbonate + water + carbon dioxide



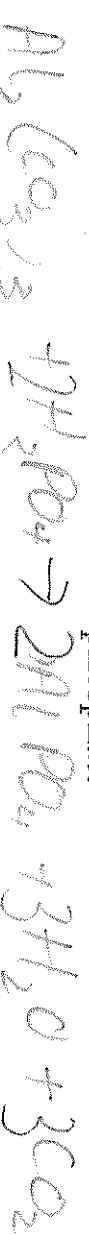
11. calcium + phosphoric acid $\rightarrow$ calcium phosphate + hydrogen gas



12. sodium + oxygen gas $\rightarrow$ sodium oxide



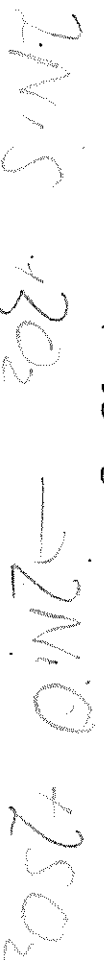
13. aluminium + phosphoric acid $\rightarrow$ aluminium + water + carbon dioxide
phosphate



14. iron (III) oxide + nitric acid $\rightarrow$ iron (III) nitrate + water



15. nickel sulfide + oxygen gas $\rightarrow$ nickel oxide + sulfur dioxide



SHEET 2

Write and balance the following equations:

1. sodium hydroxide + hydrochloric acid $\rightarrow$ sodium chloride + water



2. calcium carbonate + sulfuric acid $\rightarrow$ calcium sulfate + water + carbon dioxide



3. zinc + sulfuric acid $\rightarrow$ zinc sulfate + hydrogen gas



4. zinc hydroxide + acetic acid $\rightarrow$ zinc acetate + water



5. sodium hydrogen carbonate + hydrochloric acid $\rightarrow$ sodium chloride + water + carbon dioxide



6. calcium hydroxide + nitric acid $\rightarrow$ calcium nitrate + water



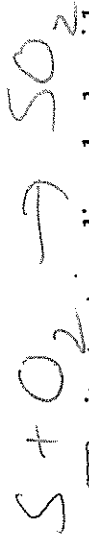
7. potassium hydroxide + nitric acid $\rightarrow$ potassium nitrate + water



8. sodium carbonate + sulfuric acid $\rightarrow$ sodium sulfate + water + carbon dioxide



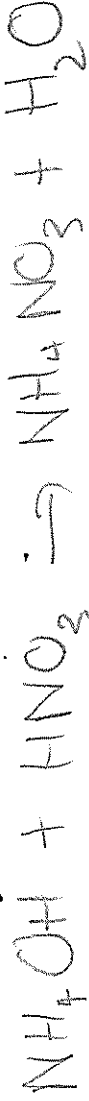
9. sulfur + oxygen gas $\rightarrow$ sulfur dioxide



10. iron (III) nitrate + sodium hydroxide $\rightarrow$ iron (III) hydroxide + sodium nitrate



11. ammonium hydroxide + nitric acid $\rightarrow$ ammonium nitrate + water



12. calcium + sulfuric acid $\rightarrow$ calcium sulfate + hydrogen gas



13. zinc oxide + hydrochloric acid $\rightarrow$ zinc chloride + water



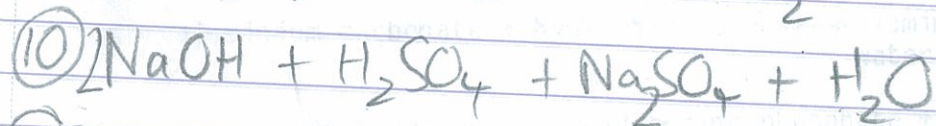
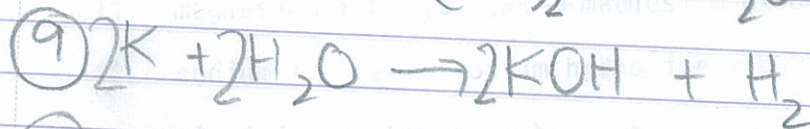
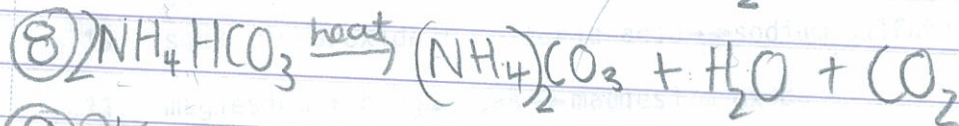
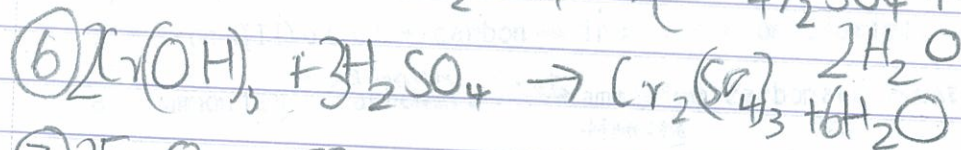
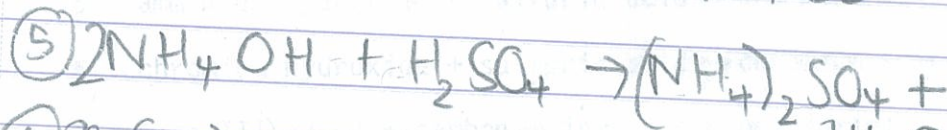
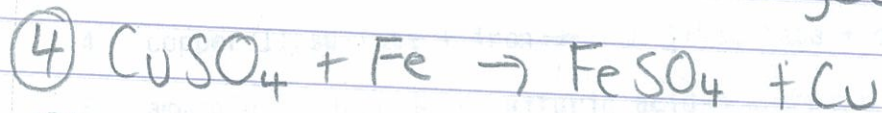
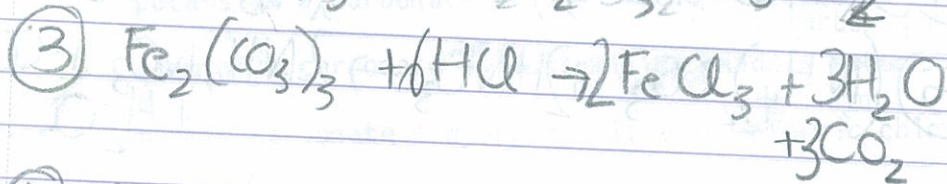
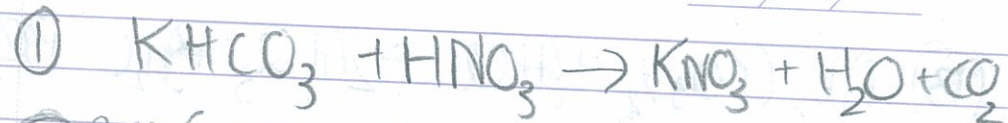
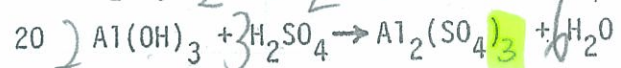
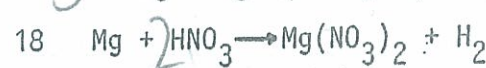
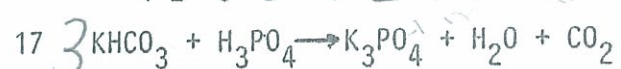
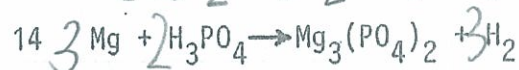
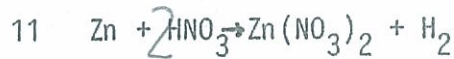
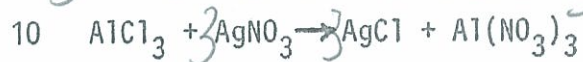
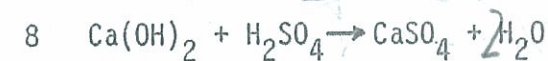
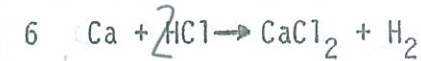
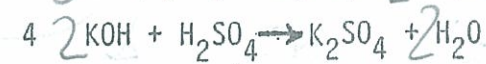
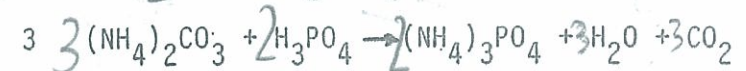
14. lead nitrate + zinc chloride $\rightarrow$ lead chloride + zinc nitrate



15. iron (III) hydroxide + sulfuric acid $\rightarrow$ iron (III) sulfate + water



Balance the following equations



+ water

carbon dioxide

+ water + CO_2

+ water

+ water

+ carbon dioxide

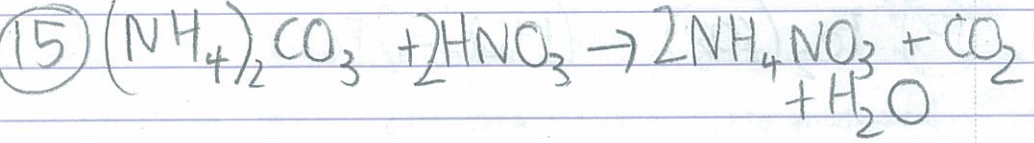
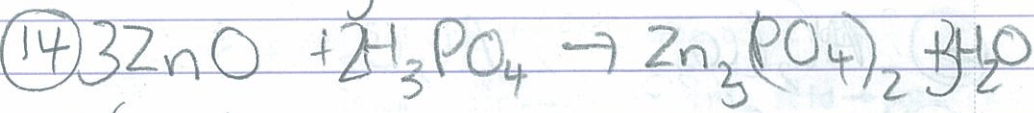
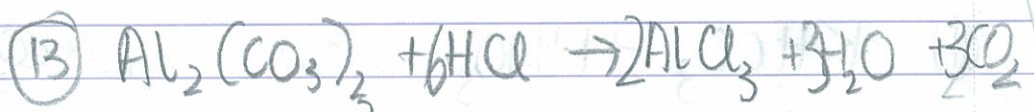
s

water

chloride + carbon dioxide

r

+ carbon



oxide + water

+ carbon dioxide

oxide + water + CO₂

oxide

oxide + water

oxide + water

oxide + carbon dioxide

oxide gas

oxide water

chloride + carbon dioxide

oxide

oxide + carbon

Balance the following equations

